

PROGRAM : COMPUTER ENGINEERING

SEM : I

YEAR : 1ST

COURSE : ENGINEERING GRAPHICS

COURSE CODE : ENG238904

**NAME OF THE FACULTY: SHARYU R. KADAM ,
MAANVI BHAGAT**

1. COURSE DETAILS

Programme: Computer Engineering/Information Technology

Semester: I/I

Course: Engineering Graphics

Course Category: DSC

Course Code: ENG238904

Duration:16 Weeks

2. LEARNING AND ASSESSMENT SCHEME

Learning Scheme				Credits	Assessment Scheme								
Actual Contact Hrs./Week			Self-Learning (SL ^) (Term Work + Assignment) (Hrs)		Paper Duration (Hrs.)	Theory (Marks)			Based on LL & TL			Based on Self Learning	Total Marks
CL	TL	LL							Practical (Marks)				
						FA-TH	SA-TH	Total	FA-PR	SA-PR	SA-OR		
2	-	2	-	2	-	-	-	-	50	-	-	-	50

CO'S

CO No.	COURSE OUTCOME	Bloom's LEVEL
1.	Use different CAD commands for drawing.	Apply
2.	Construct engineering scales, curves and Loci of points.	Understand
3.	Draw orthographic projection and isometric projection of given engineering components.	Remember
4.	Draw various types of planes.	Remember

Sr. No.	TOPIC/Sub-topic	Hours	Marks	COs
1	Drawing Instruments and their uses 1.1. Letters and numbers (single stroke vertical), 1.2. Convention of lines and their applications, 1.3. Scale (reduced, enlarged & full size) plain scale and diagonal scale, Sheet layout, 1.4. Introduction to Computer Aided Drafting (Basic draw and modify Command), Geometrical constructions. 1.4. Introduction to Computer Aided Drafting (Basic draw and modify Command), Geometrical	05		CO1
2	Engineering curves & Loci of Points 2.1. To draw an ellipse by: Directrix and focus method, Arcs of circle method 2.2. To draw a parabola by: Directrix and focus method, Rectangle method, 2.3. To draw a hyperbola by: Directrix and focus method, passing through given points with reference to asymptotes, Transverse Axis and focus method, 2.4. To draw involutes of circle & polygon 2.5. To draw a cycloid, epicycloids, hypocycloid, 2.6. To draw Helix & spiral, 2.7. Loci of Points: Loci of points with given conditions and examples related to simple mechanisms.	10		CO2
3	Planes 3.1. Lines inclined to one reference plane only and limited to both ends in one quadrant, 3.2. Projection of simple planes of circular, square, rectangular, rhombus, pentagonal, and hexagonal, inclined to one reference plane and perpendicular to the other.	05		CO4
4	Orthographic projections 4.1. Introduction to Orthographic projections, 4.2. Conversion of pictorial view into Orthographic Views (First Angle Projection Method Only), 4.3. Dimensioning technique as per SP-46	06		CO3
5	Isometric projection 5.1. Isometric scale, Conversion of orthographic views into isometric View/projection (Simple objects) 5.2. Projection of Straight Lines and Planes (First Angle Projection Method only)	06		CO3
TOTAL		32		

LIST OF PRACTICALS/ASSIGNMENTS/ TUTORIALS/DRAWINGS

Term Work consists of Journal containing minimum no of –10 Experiments/drawings.

Sr. No.	Title of Experiment/Assignment/Exercise/Tutorial/Drawings	Approx. Hrs required	CO
1.	Draw horizontal, vertical, 30 degree, 45 degree, 60 and 75 degrees lines, different types of lines.	02	CO1
2.	Write alphabets and numerical.	02	CO1
3.	Draw basic 2D shapes like: line, rectangle circle, ellipse, arc polygon using computer aided drafting approach	02	CO1
4.	Use computer aided drafting approach to apply modifications on 2D shapes.	02	CO1
5.	Draw different types of scales	02	CO1
6.	Draw different types of Dimensions	02	CO1
7.	Draw different types of Engineering Curves	04	CO2
8.	Draw Loci of Point	02	CO2
9.	Draw Projection of Points and Straight Lines	02	CO3
10.	Draw Projection of Planes.	02	CO3
11.	Draw a problem on orthographic projections using first angle method	02	CO4
12.	Draw Isometric Projection.	02	CO4
13.	Demonstrate 3D printing process	02	CO4
14.	Correlate ancient Indian sculptures, Indian temples, Monuments, etc. with Engineering Graphics	02	All COs
	TOTAL	30	

CH- 1

Drawing Instruments and their uses

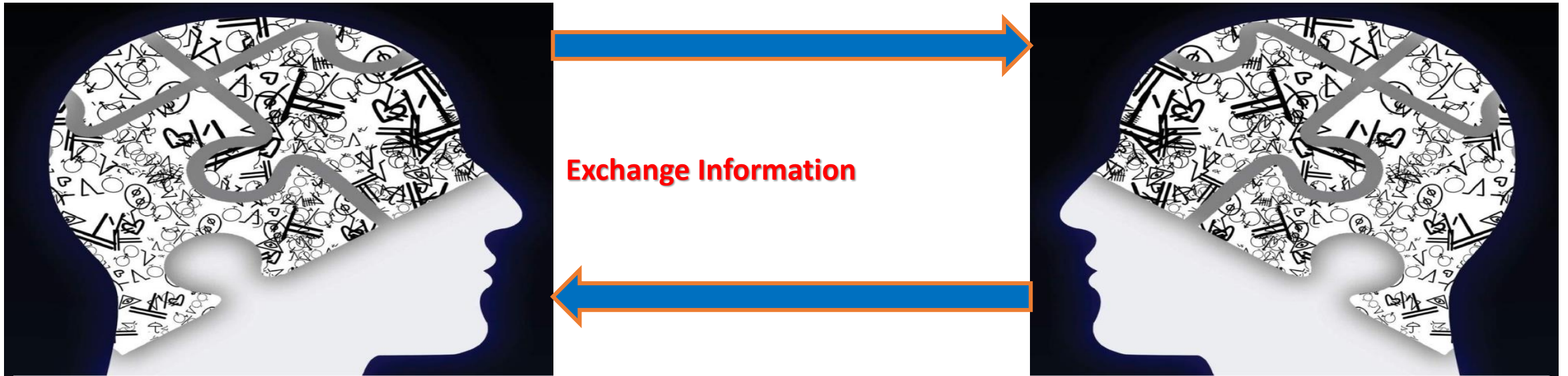
Engineering Drawing

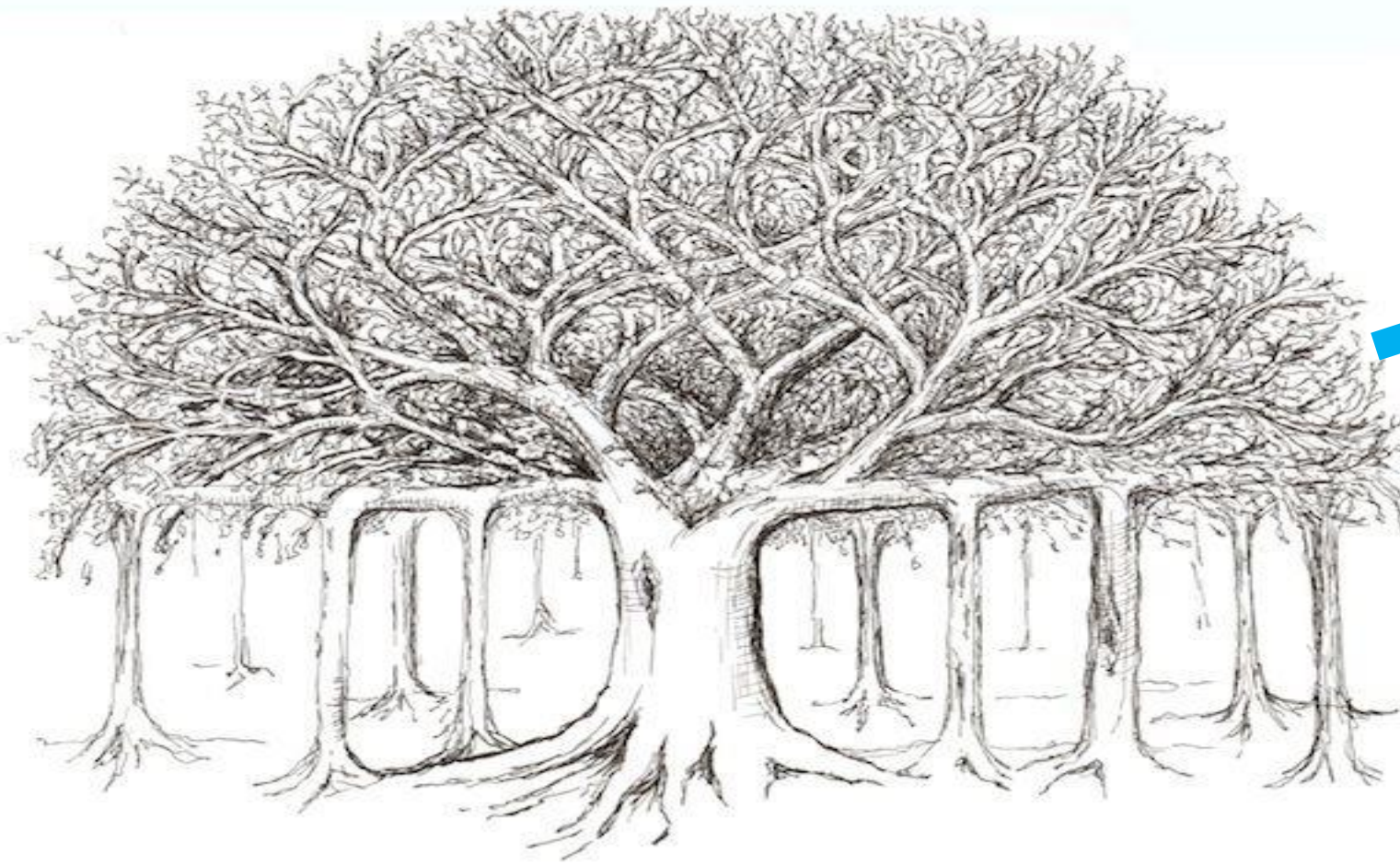
➤ Technical drawing



Engineering Drawing

➤ Graphical Language





Engineering Drawing



Artistic Drawing

Why this banyan tree drawing is not engineering drawing ?

➤ Engineering is the profession in which a knowledge of mathematical and natural science gained by study, experience and practice is applied with judgment to develop product.

- In a process of product development, two steps are involved:

- Product specification

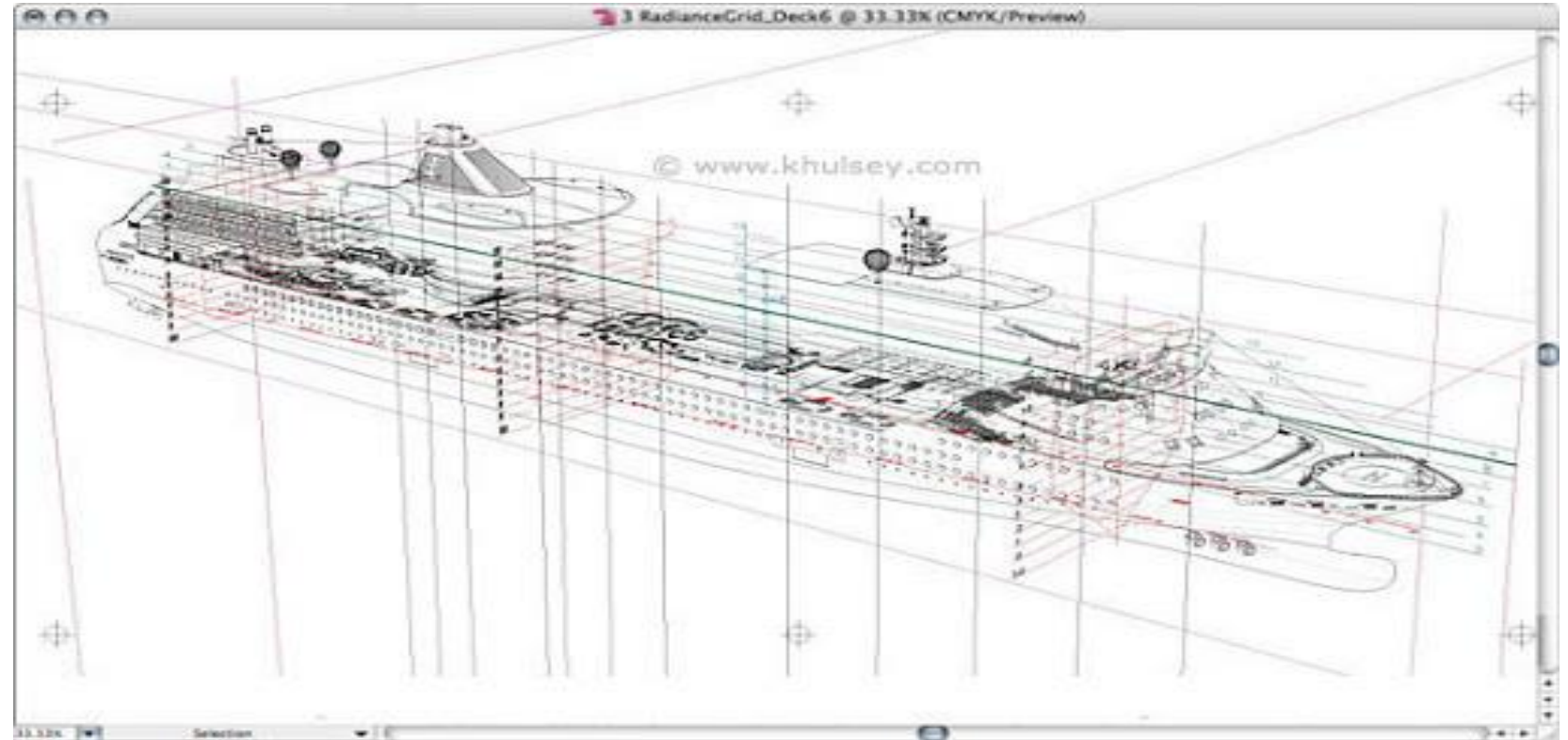
- Product drawing

➤ Although the works of artists (or photography and other method of reproduction) may provide pictorial representation, but they cannot serve as engineering description.

Engineers must be able to create idea sketches, calculate stresses, analyze motions, size parts, specify materials and production methods, make design layouts and supervise the preparation of drawings and specifications that will control the numerous details of product manufacture, assembly and maintenance

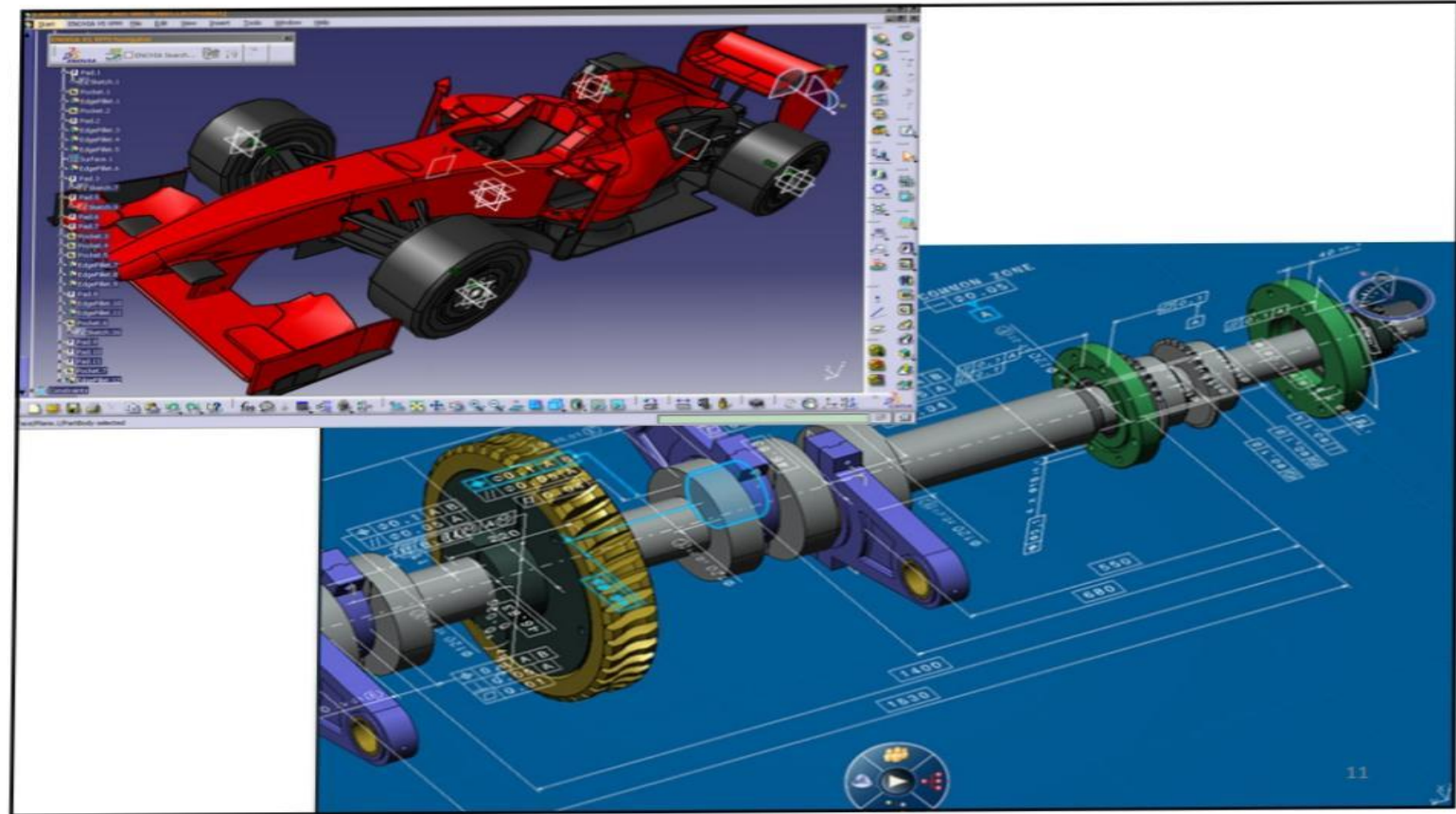
Applications of Engineering Graphics :

➤ Ships



Applications of Engineering Graphics :

➤ Automobiles :



Applications of Engineering Graphics :

➤ Construction :



Prerequisite For Engineering Drawing

Table 1.1 Recommended Sizes of Drawing Sheets and Drawing Boards

<i>Drawing Sheet (IS 10711:2001)</i>		<i>Drawing Board (IS 1444:1989)</i>	
<i>Designation</i>	<i>Size (mm) Length × Width</i>	<i>Designation</i>	<i>Size (mm) Length × Width</i>
A0	1189 × 841	D0	1270 × 920
A1	841 × 594	D1	920 × 650
A2	594 × 420	D2	650 × 470
A3	420 × 297	D3	500 × 350
A4	297 × 210		

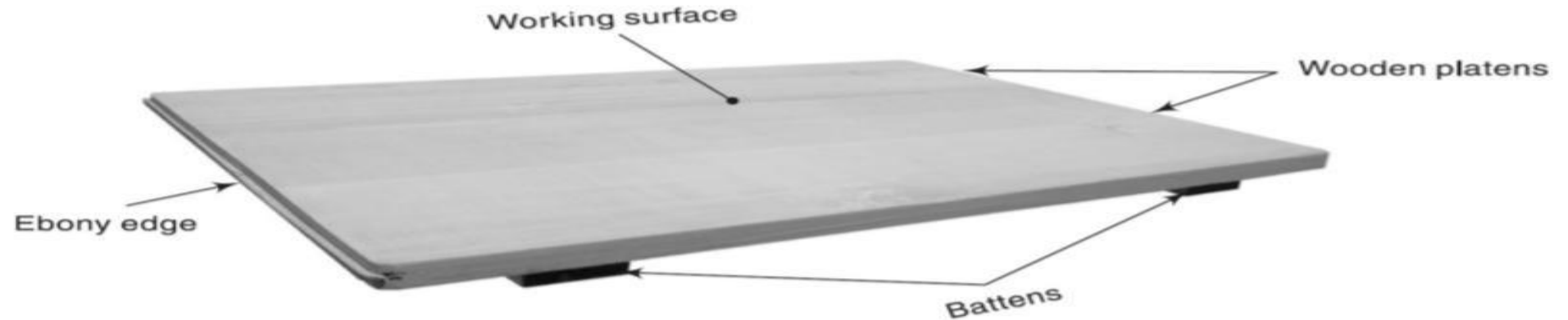
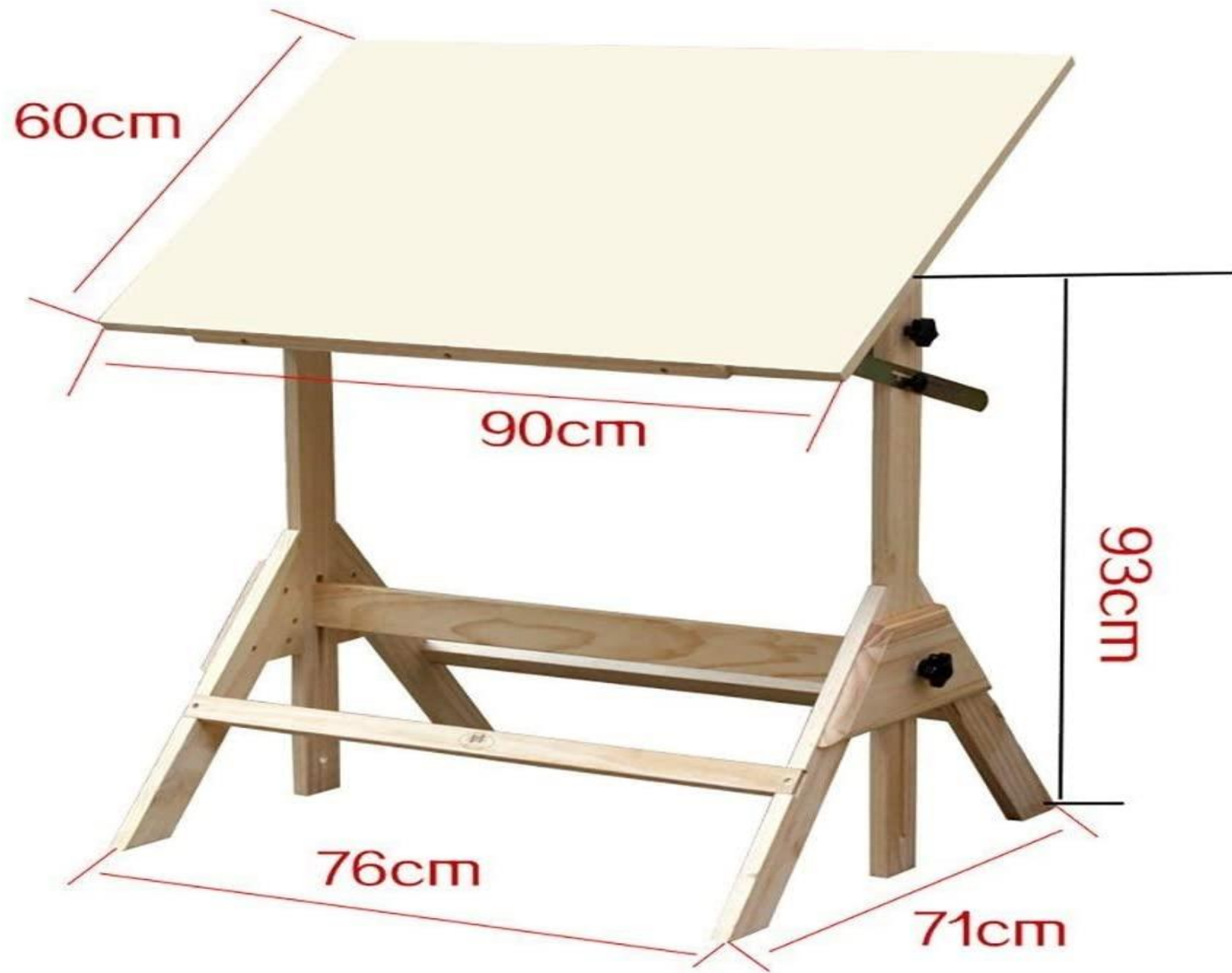
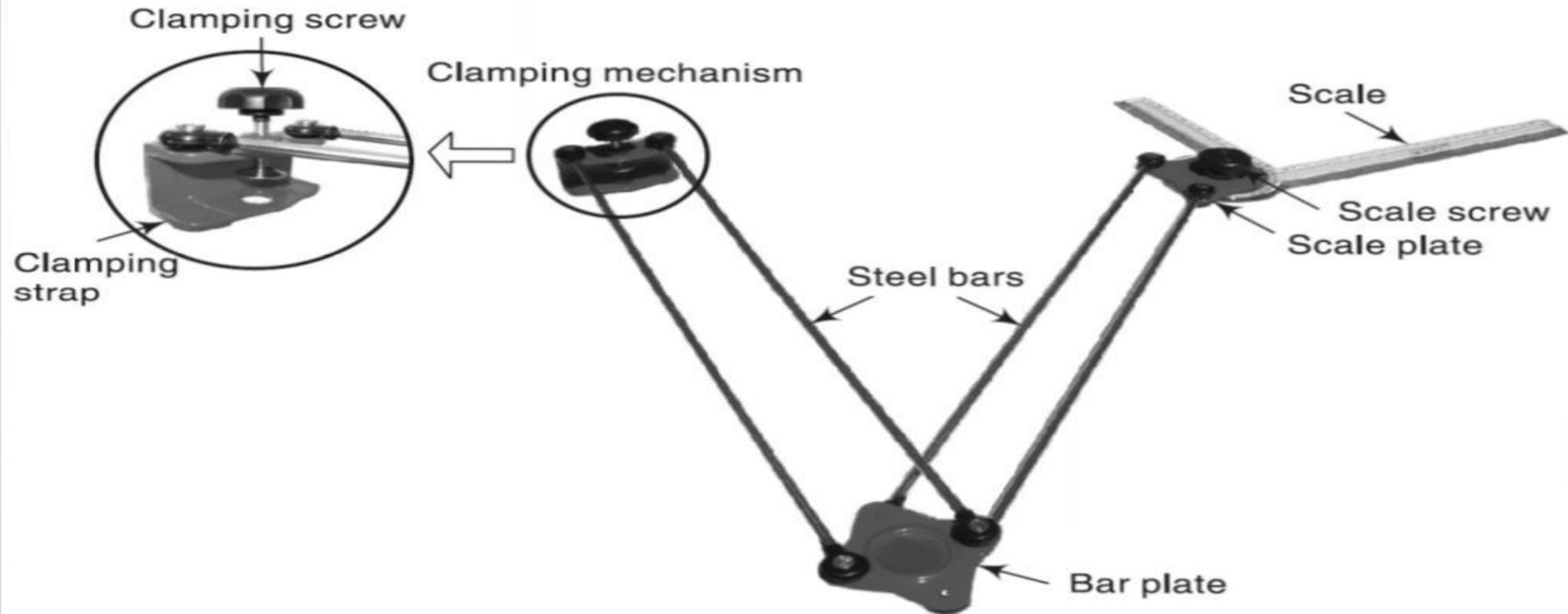
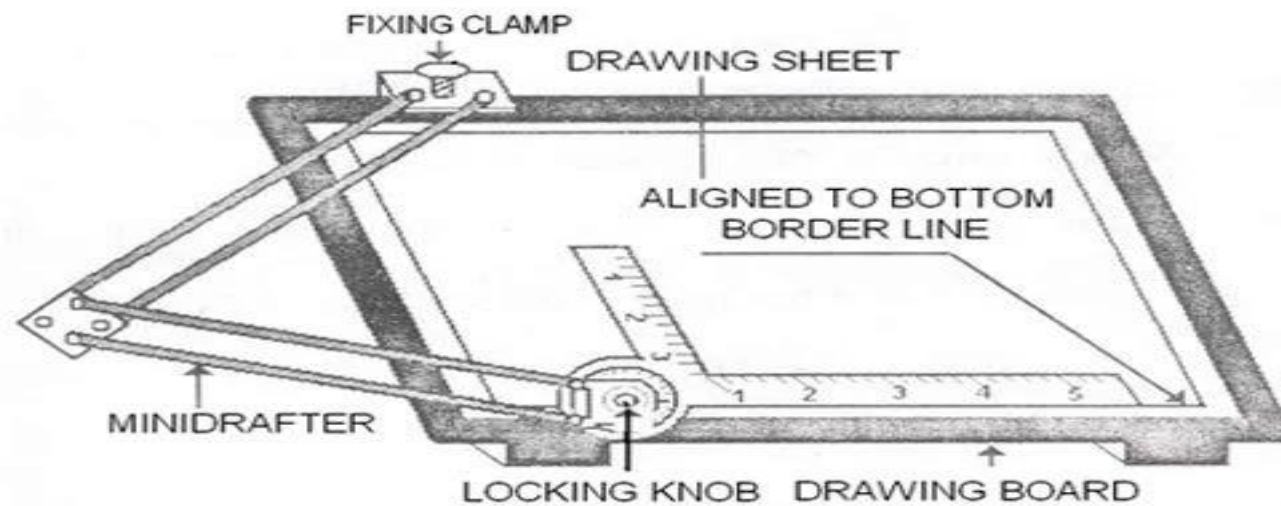
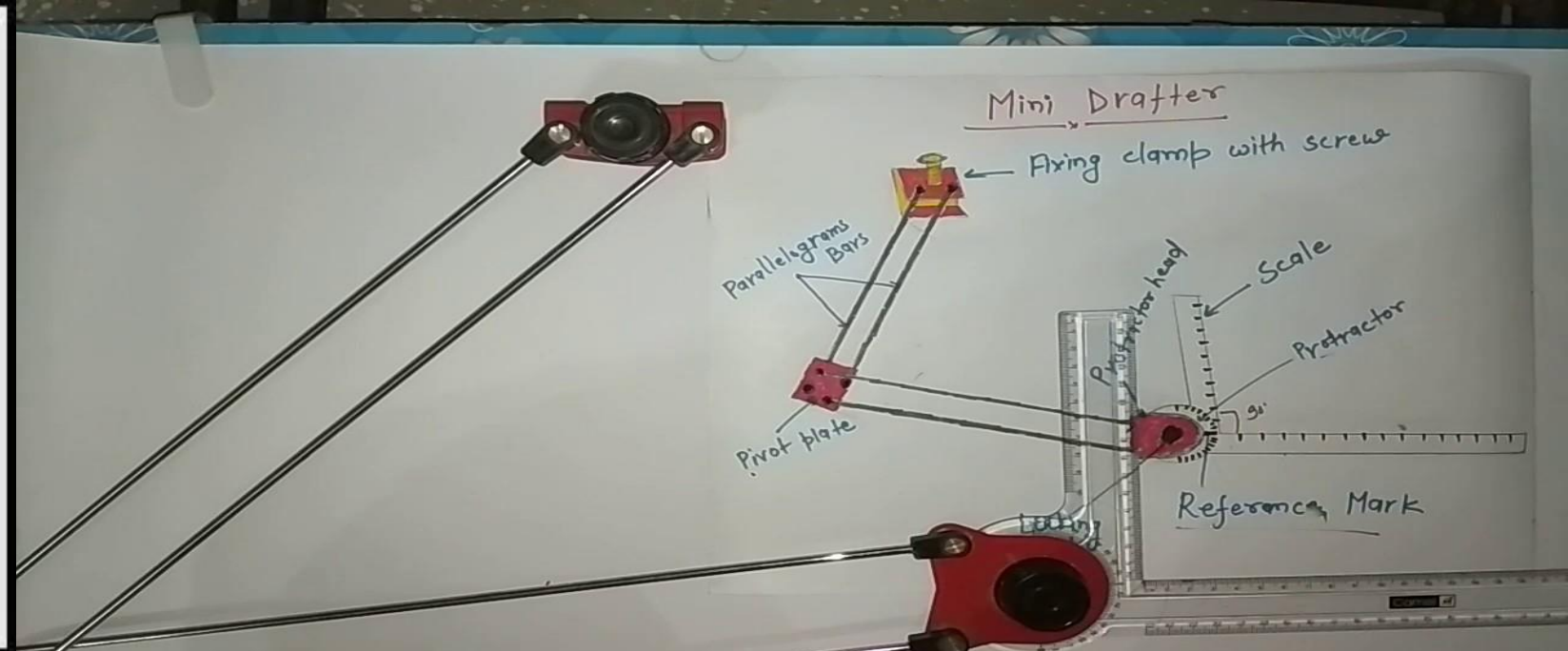


Fig. 1.2 Drawing board

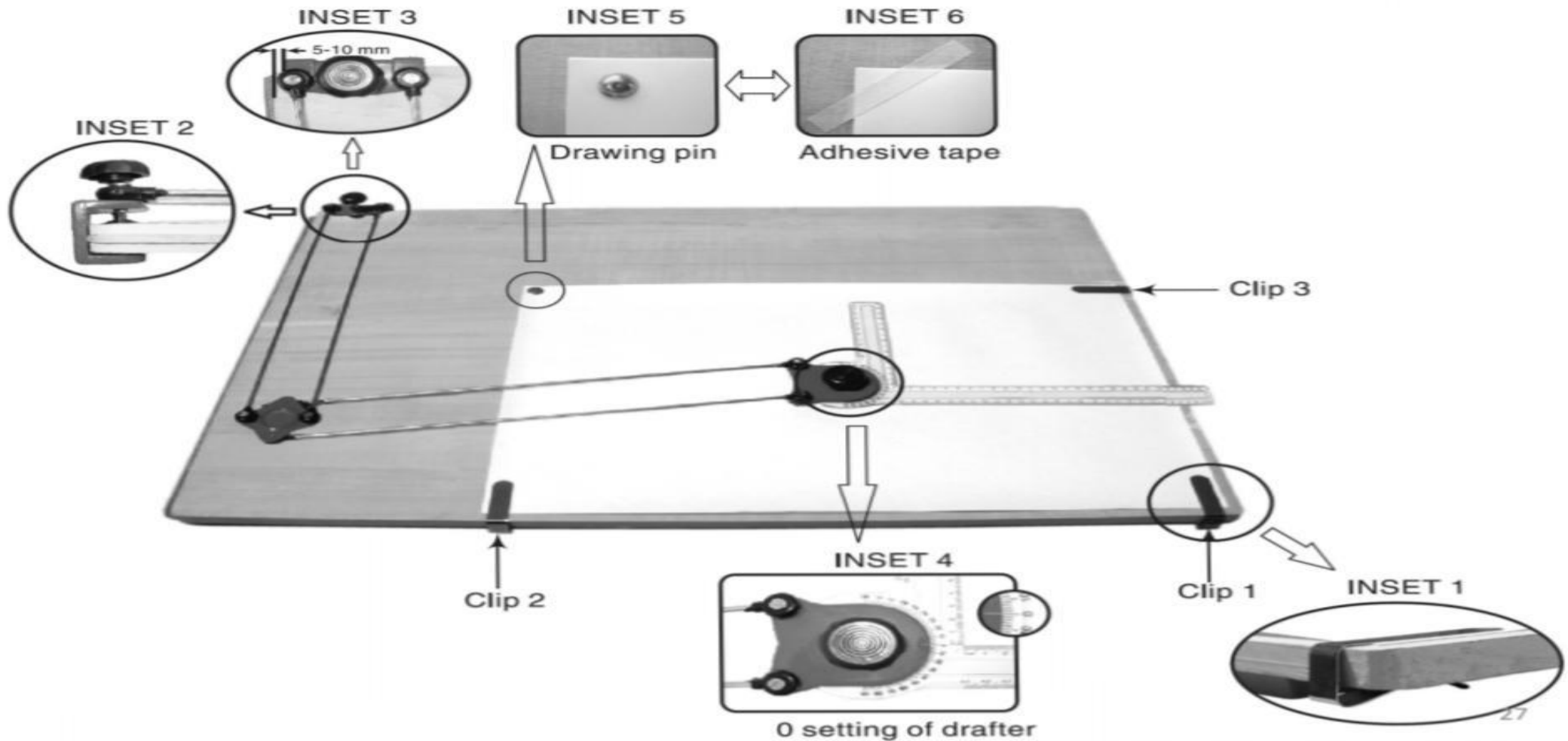


Prerequisite For Engineering Drawing





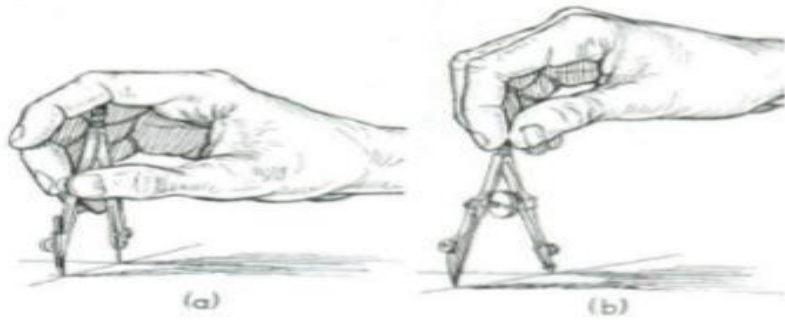
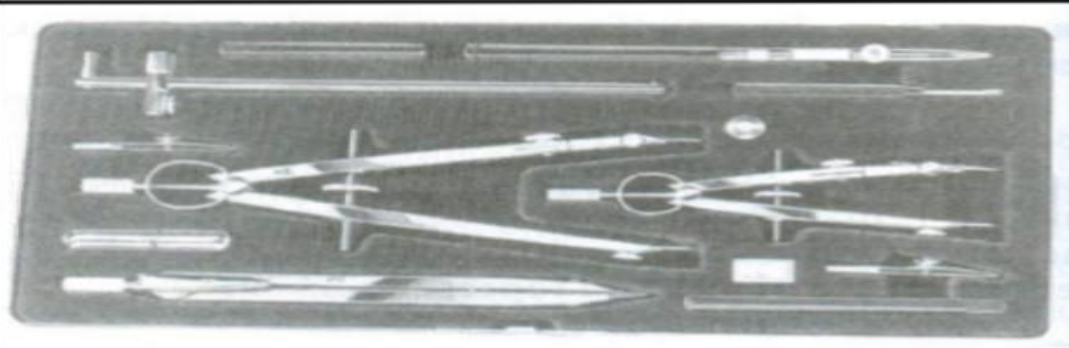
Prerequisite For Engineering Drawing



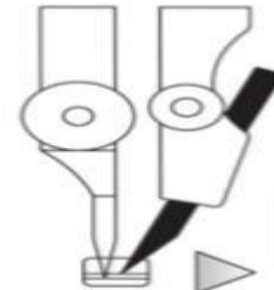
➤ Drawing Sheet

Table 1.2 Standard Sizes of Drawing sheets as per BIS

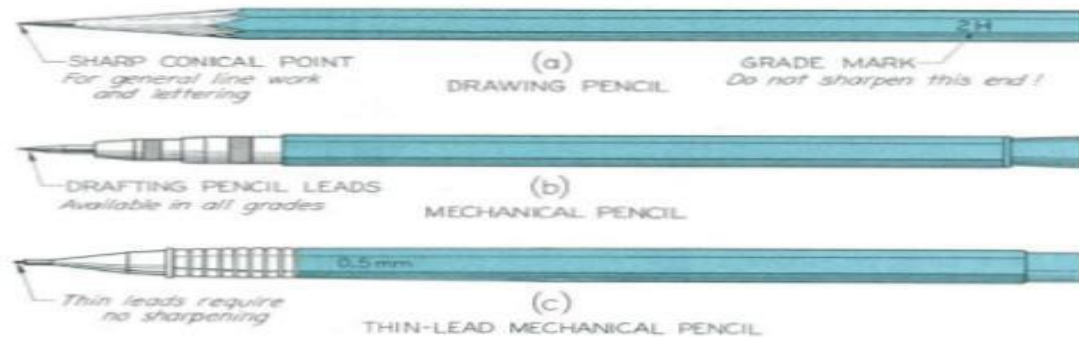
Designation	Size (mm)
A0	841 x 1189
A1	594 x 841
A2	420 x 594
A3	297 x 420
A4	210 x 297



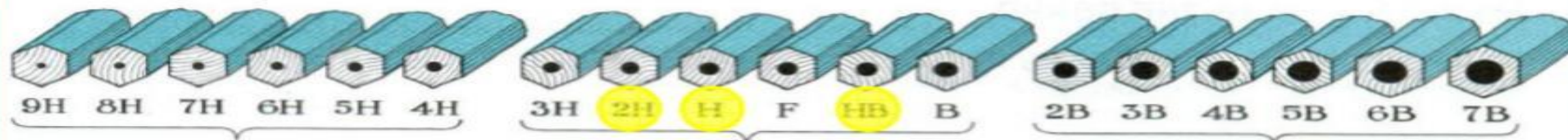
Lead sticks



Prerequisite For Engineering Drawing



Lead Grade Sheet



Hard

The hard leads in this group (left) are used where extreme accuracy is required, as on graphical computations and charts and diagrams. The softer leads in this group (right) are sometimes used for line work on engineering drawings, but their use is restricted because the lines are apt to be too light.

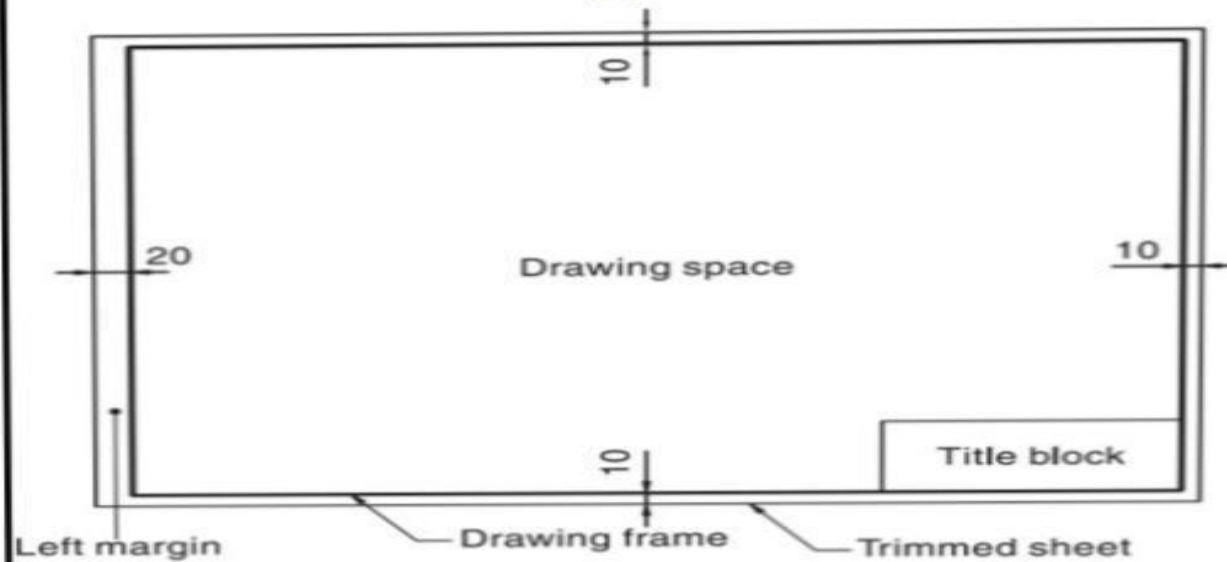
Medium

These grades are for general purpose work in technical drawing. The softer grades (right) are used for technical sketching, for lettering, arrowheads, and other free-hand work on mechanical drawings. The harder leads (left) are used for line work on machine drawings and architectural drawings. The H and 2H leads are widely used on pencil tracings for reproduction.

Soft

These leads are too soft to be useful in mechanical drafting. Their use for such work results in smudged, rough lines that are hard to erase, and the lead must be sharpened continually. These grades are used for art work of various kinds, and for full-size details in architectural drawing.

Prerequisite For Engineering Drawing



Drawing margins and location of title block

65	10	NAME	
	10	INSTITUTE	
	10	ROLL No.	TUT. GROUP:
	10	SHEET No:	DATE:
	10	TUTORIAL TITLE	
20		80	
160		29	

Lettering

- Lettering means
 - Writing of titles, dimensions, notes and other important particulars
- Types
 - Single stroke letters
 - Double stroke letters
- Single Stroke
 - Thickness of the line of the letter should be such as is obtained in one stroke of the pencil.
 - Does not mean that the letter should be made in one stroke without lifting the pencil.

Basic Strokes

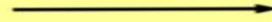
Straight



Slanted



Horizontal



Curved

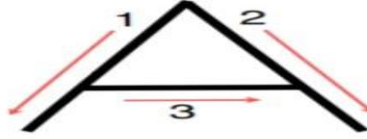


Examples

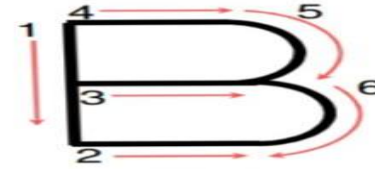
"I" letter



"A" letter



"B" letter



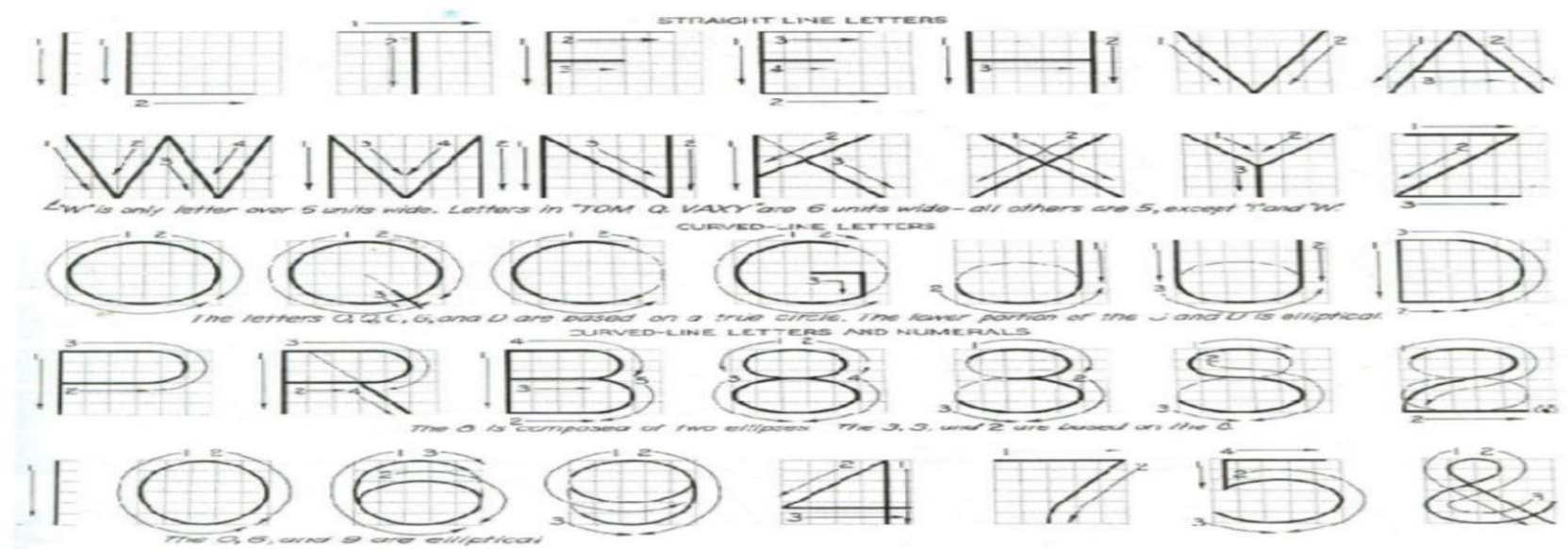
31

Height and Width of Letters

- BIS (SP 46: 2003) has recommended the heights of letters as:
 - 1.8 mm, 2.5 mm, 3.5 mm, 5 mm, 7 mm, 10 mm, 14 mm and 20 mm.
 - Large-sized letters are used for main titles and headings,
 - Medium-sized letters for subtitles and important notes and
 - Small-sized letters for dimensions and general notes.
 - The height-to-width ratio varies from letter to letter. Most of the letters follow the ratio 7 : 5 or 7 : 6.

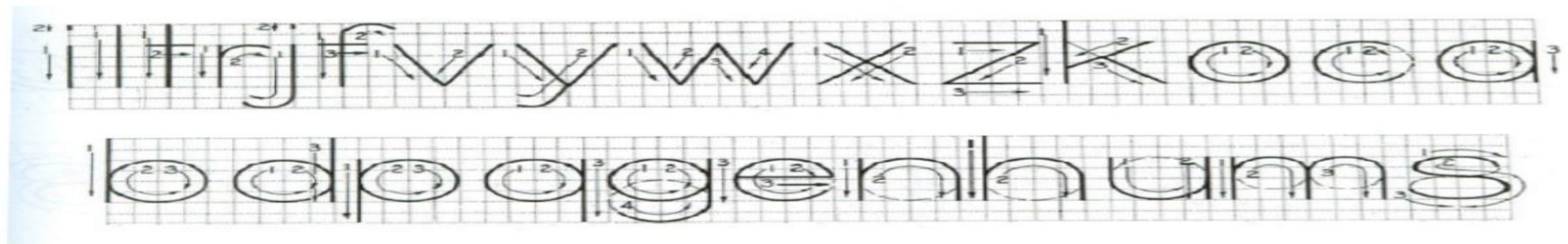


32



Vertical Capital Letters and Numerical

33



Vertical Lowercase Letters

Double Stroke Lettering

- More thickness is given to single stroke letters, it is known as double stroke or gothic letters.

A B C D

Lettering Rules

1. **Draw letters as simple as possible. Artistic or cursive lettering should be strictly avoided.**
2. **Draw letters symmetrical about the vertical axis or horizontal axis. Asymmetric letters like, F, R, Z, 4, etc., may be drawn as they are.**
3. **Round-off the sharp corners wherever necessary, e.g., D, P, S, etc.**
4. **Draw all letters legible and uniform.**
5. **The height of all the letters in one line should be the same.**
6. **Use single stroke vertical CAPITAL letters as much as possible.**

Graphic Language in Engineering Drawing

- “Engineering drawing” or “blueprint” uses *lines* to represent the *features* of an object.
- Features of an object are *surface* (include *plane*) and *edge*.

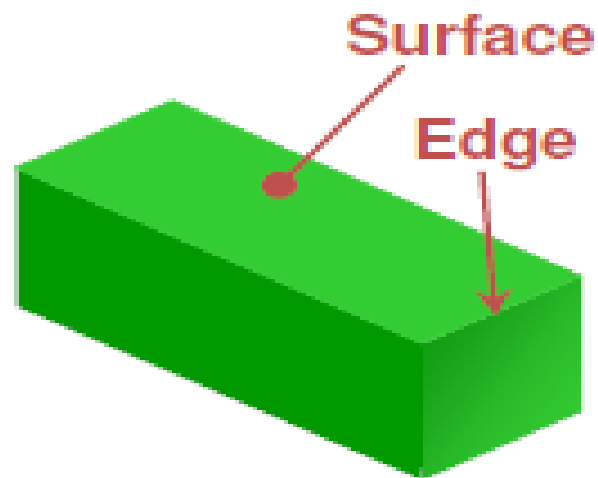
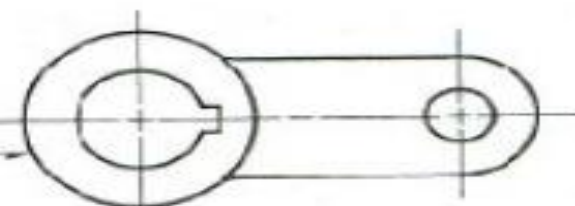


Illustration	Application
Thick 	Outlines, visible edges, surface boundaries of objects, margin lines
Continuous thin 	Dimension lines, extension lines, section lines leader or pointer lines, construction lines, boarder lines
Continuous thin wavy 	Short break lines or irregular boundary lines – drawn freehand
Continuous thin with zig-zag 	Long break lines
Short dashes, gap 1, length 3 mm 	Invisible or interior surfaces
Short dashes 	Center lines, locus lines Alternate long and short dashes in a proportion of 6:1,
Long chain thick at end and thin elsewhere 	Cutting plane lines

Lines

Visible line

THICK
Approx. width 0.6 mm (.024")



Hidden line

0.0 mm (.003") THIN
Approx. width 0.3 mm (.012")



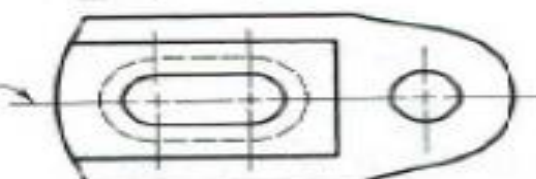
Section line

THIN



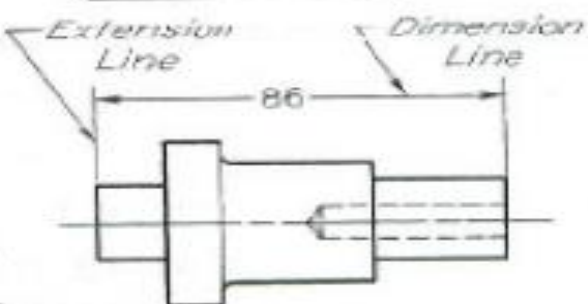
Center line

19 - 38 mm (.75 - 1.50") THIN
3.2 mm (.12")
1.5 mm (.06") THIN



Dimension line,
Extension line

90.5 THIN



Leaders

1.6 mm (.06") THICK 6.4 mm (.25")

Cutting-plane
or
Viewing-plane
lines

3.2 mm (.12")
1.6 mm (.06") THICK 19 - 38 mm (.75 - 1.50")



SCALES :

A **scale** is defined as the ratio of the linear dimensions of the object as represented in a **drawing** to the actual dimensions of the same.

Some objects like buildings, machines etc are big



some objects are very small like parts used in wrist watches



- Some Drawing of above is required to be drawn proportionally reduced or enlarged.
- The proportion by which we are enlarging or reducing actual length of the object on drawing is called as **scale**.

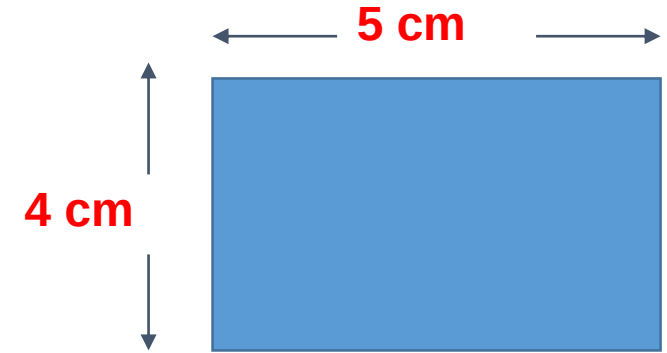
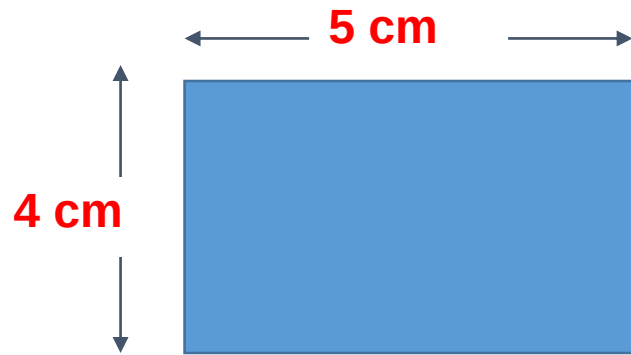
Scale Size :

1.Full size.

2.Reduced size.

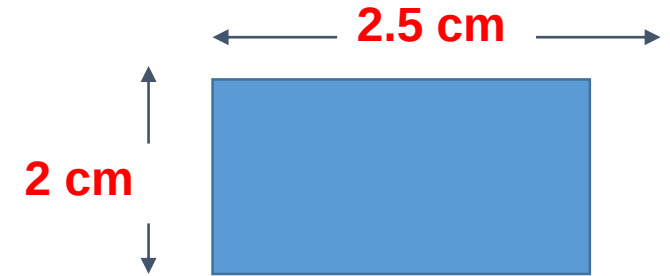
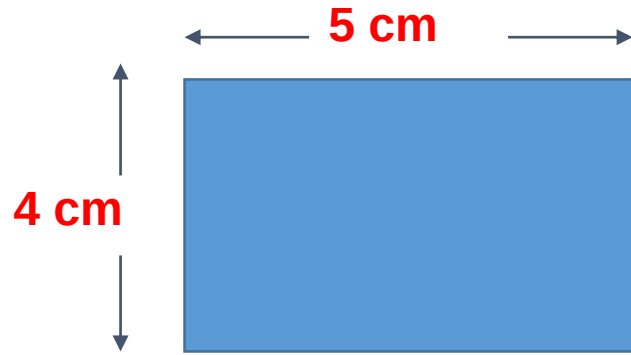
3.Enlarged size.

1. Full size



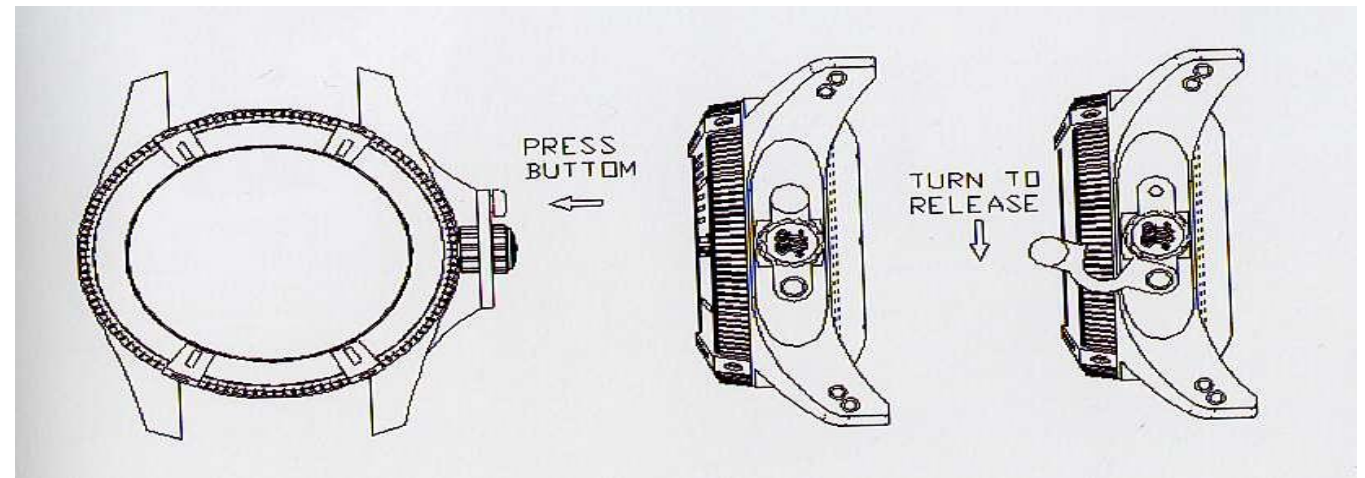
- The scale in which objects are drawn with actual measurement is called as full size scale.
- It is written in ratio **1:1**

2. Reduced size



- The scale in which objects are drawn with reduced proportion is called as reduced size scale.
- It is written in ration **1:2 , 1:5, 1:10, 1:20**

2. Enlarged size



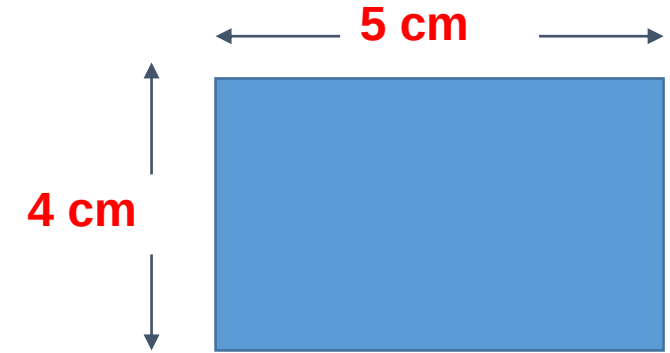
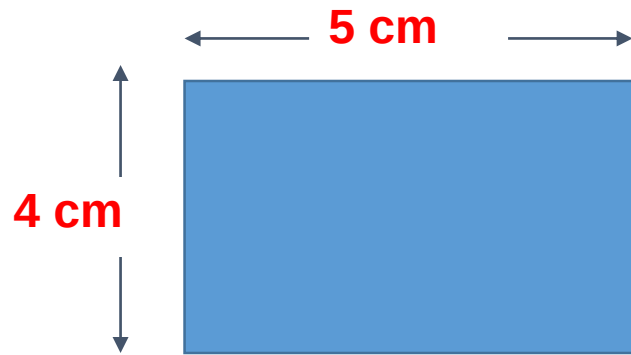
- The scale in which objects are drawn with increased proportion is called as reduced size scale.
- It is written in ration **2:1 , 5:1, 10:1, 20:1**

Representative Fraction (R.F.)

It is ratio of the length of the object on the drawing to the actual length of the object

$$\text{R.F.} = \frac{\text{Length of object on the drawing}}{\text{Actual length of object}}$$

1. Full size

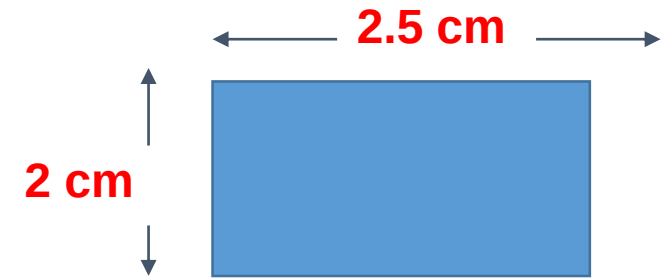
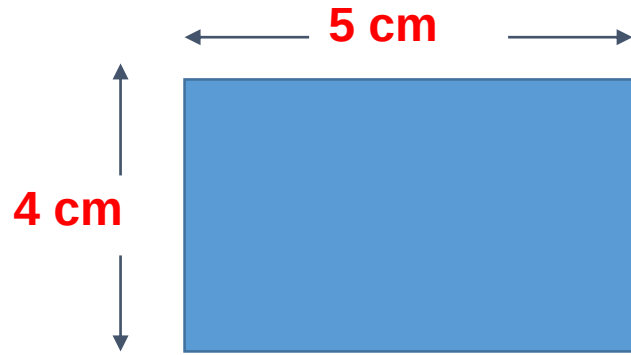


R.F. = 1



- The scale in which objects are drawn with actual measurement is called as full size scale.
- It is written in ration **1:1**

2. Reduced size

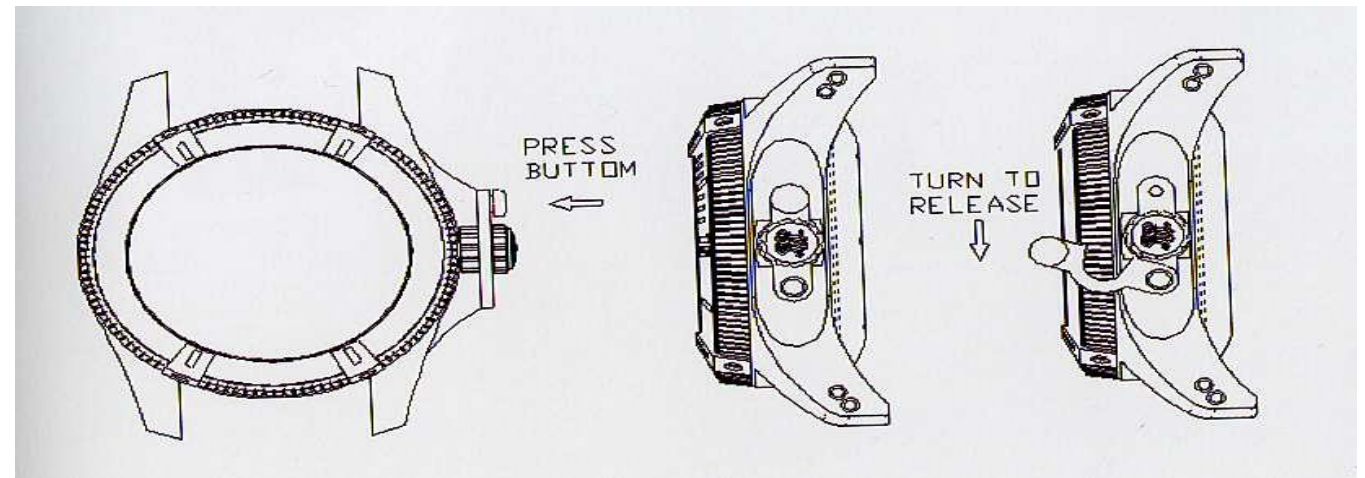


R.F. < 1



- The scale in which objects are drawn with reduced proportion is called as reduced size scale.
- It is written in ration **1:2 , 1:5, 1:10, 1:20**

2. Enlarged size



R.F. > 1

- The scale in which objects are drawn with increased proportion is called as reduced size scale.
- It is written in ration **2:1 , 5:1, 10:1, 20:1**

Type Of Scale :

1.Plain Scale.

2.Diagonal Scale.

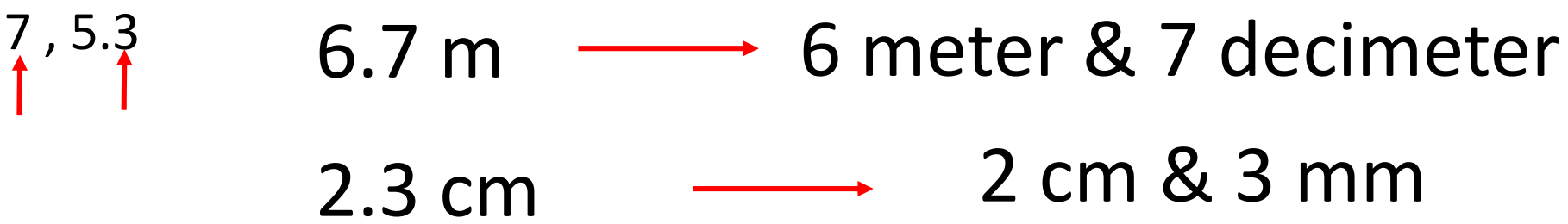
Metric Measurements:

mm	=	Millimeter
cm	=	centimeter
dm	=	decimeter
m	=	meter
dam	=	decameter
hm	=	hectometer
km	=	kilometer

1 cm	=	10 mm
1 dm	=	10 cm
1 m	=	10 dm
1 dam	=	10 m
1 hm	=	10 dam
1 km	=	10 hm
1 Yard	=	3 Feets
1 feet	=	12 inches
1 inch	=	2.54 cm

Note : Always calculate in cm

Plain Scale

- This **scale** is used when very small distances are to be accurately measured for example 0.1 mm
- When measurements are required up to one decimal place (i.e. 2 unit)
- Example : 6.7 , 5.3


6.7 m	→	6 meter & 7 decimeter
2.3 cm	→	2 cm & 3 mm

For constructing plain scale we must know:

1. Its representative fraction i.e. (R.F.) (given or calculate)
2. Maximum length to be measured (given)
3. The units to be shown, for example, millimeters and centimeters(given)

Steps for constructing plain scale

1. Calculate the length of scale.

$$\text{length of scale} = \text{R.F} \times \text{length to measured}$$

2. Draw two horizontal lines parallel to each other and some distance apart so that they give appearance as a rule. Join the ends by vertical lines.

(1. Length for horizontal parallel lines = length of scale which is calculated previously 2. assume suitable length)

3. Divide horizontal lines into required number of equal parts and (depend upon length to measured or read upto)

4. draw perpendicular lines at these division points.

5. Mark 'zero' at the end of first division.

6. sub-divide only first main part to the left of zero into the required number of sub-division according to unit.

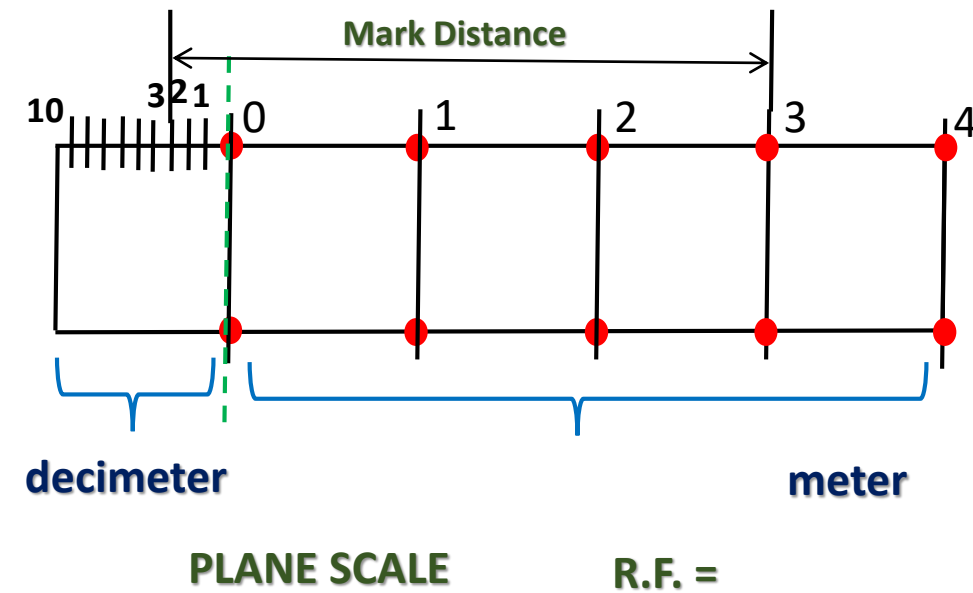
$$1 \text{ m} = 10 \text{ decimeter}$$

Hence we need to divide first part into 10 equal numbers

7. Sub-divisions are numbered to the right, starting from zero.

8. Mark the given distance on scale

9. name of the scale and R.F. should be mentioned below scale



Example : 3.2 m means 3 meter and 2 decimeter

Problem 1 :

Construct a plain scale with R.F. **1:40** to read up to **5 meters** in **meters** and **decimeters**. Mark **3.4 m** on scale

Given data : R.F. = 1/40 , maximum length = 5 m

1. Calculate the length of scale.

length of scale = R.F X length to measured
= 1/40 X 5 m
= 0.125 m
= 0.125 X 100 = **12.5 cm**

2. Draw two horizontal lines of 12.5 cm parallel to each other and 2 cm distance apart. Join the ends by vertical lines.

3. Divide horizontal lines into **required** number of **equal parts** (depend upon length to measured or read up to) according to given data for above example maximum length **or read up to is 5 m**, hence divide parallel lines into **5 equal parts**.
Size of individual part = 12.5/5 = **2.5 cm**

4. Mark 'zero' at the end of first division.

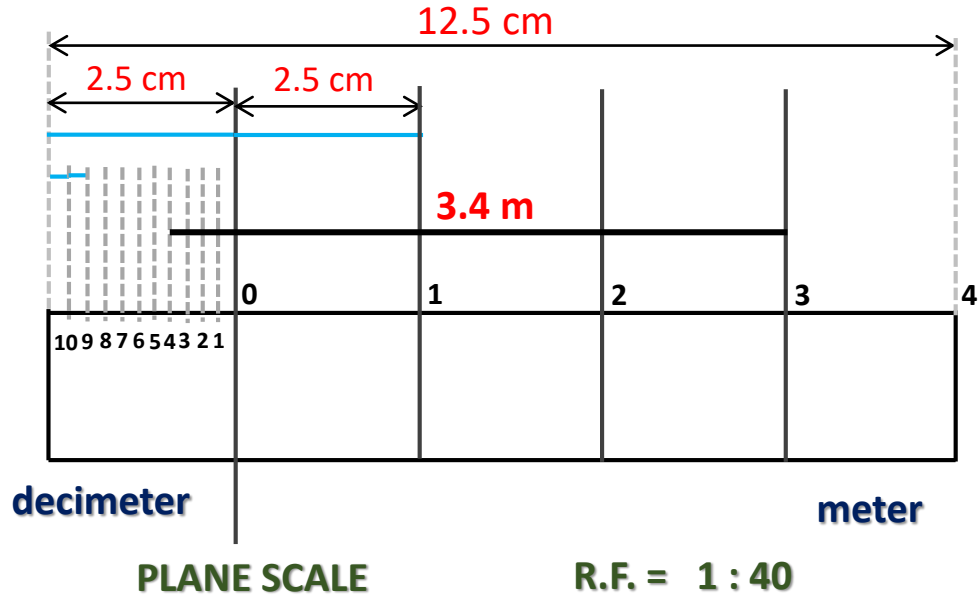
5. Sub-divide only first main part to the left of zero into the **required** number of sub-division according to unit. In above problem given units are **meter** and **decimeter** and **1 meter = 10 decimeter** hence we have to divide first main part into **10** number of **equal parts**.
Size of individual sub part = 2.5/10 = **0.25 cm**

6. Sub-divisions are numbered to the right, starting from zero.

7. Assign units for parts.

8. Mark given distance on scale. For above problem distance = **3.4 m**

9. Name of the scale and R.F. should be mentioned below scale



Problem 2 :

Construct a plain scale to show meters when 1 centimeter represents 4 meters and long enough to measure up to 60 meters.
Find the R.F. and mark on it a distance of **43 meters**.

Given data : R.F. = 1 cm is to 4 meters , maximum length = 60 m

Units : meter and centimeter

R.F. = $\frac{1 \text{ cm}}{4 \text{ m}} = \frac{1 \text{ cm}}{4 \times 100 \text{ cm}} = \frac{1 \text{ cm}}{400 \text{ cm}} = \frac{1}{400}$

1. Calculate the length of scale.

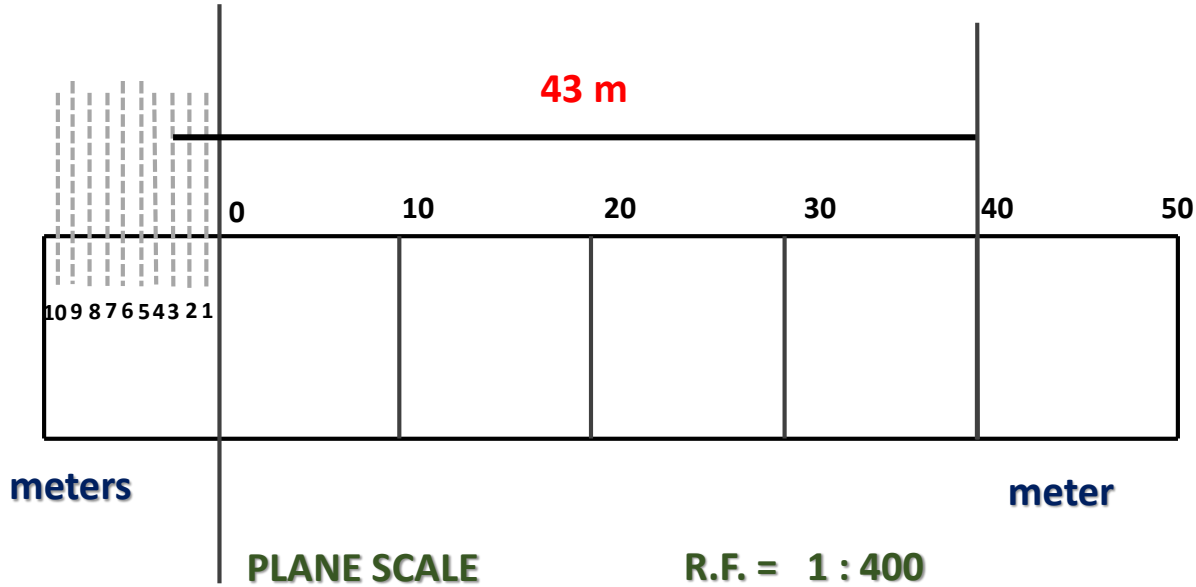
length of scale = R.F X length to measured
= $\frac{1}{400} \times 60 \text{ m}$
= 0.15 m
= $0.15 \times 100 = 15 \text{ cm}$

2. Draw two horizontal lines of 15 cm parallel to each other and 3 cm distance apart. Join the ends by vertical lines.

3. Divide horizontal lines into **required** number of **equal parts** (depend upon length to measured or read up to)
according to given data for above example maximum length **or read up to is 60 m**, hence divide parallel lines into **6 equal parts**.

Size of individual part = $15/6 = 2.5 \text{ cm}$

4. Sub-divide only first main part to the left of zero into the **10** number of sub-division.



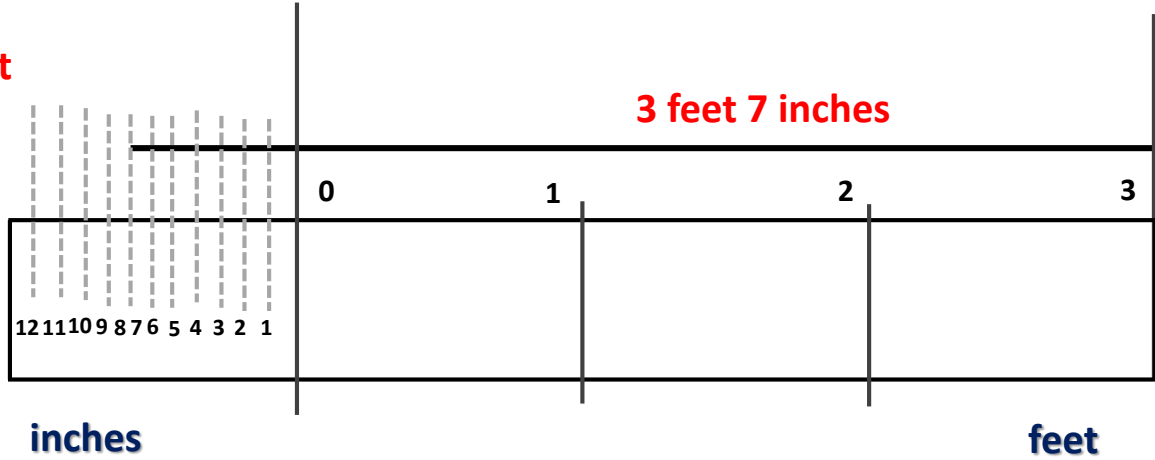
Problem 3 :

Construct a plain scale with R.F. **1:8** to show **feet** and **inches** and long enough to **measure up to 4 feet**, show a distance of **3 feet and 7 inches** on your scale.

Given data : R.F. = 1/8 , maximum length = 4 feet

1. Calculate the length of scale.

length of scale = R.F X length to measured
= 1/8 X 4 feet = 1/8 X 4 X 12 inches
= 6 inches
= 6 X 2.54 cm
= **15.24 cm**



2. Draw two horizontal lines of 15.24 cm parallel to each other and 3 cm distance apart. Join the ends by vertical lines.

PLANE SCALE **R.F. = 1 : 8**

3. Divide horizontal lines into **required** number of **equal parts** (depend upon length to measured or read up to) according to given data for above example maximum length **or read up to is 4 feet**, hence divide parallel lines into **4 equal parts**.
Size of individual part = 15.2/4 = **3.8 cm**

4. Sub-divide only first main part to the left of zero into the **12** number of sub-division. In above problem given units are **feet** and **inches** and **1 feet = 12 inches** hence we have to divide first main part into **12** number of **equal parts**.
Size of individual sub part = 3.8/12 = **0.32 cm**

Problem 4 :

Construct a plain scale with R.F. **1:40** and maximum length is **4 meters**. Mark **2.7 m** on scale where units are in **meters** and **decimeters**.

Given data : R.F. = $1/40$, maximum length = 5 m

Problem 5 :

Construct a plain scale with R.F. **1:50000** to show **kilometers** and **hectometers** and long enough to measure up to **7 kilometers**. Measure a distance of **46 hectometers** on your scale.

Given data : R.F. = $1/50000$, maximum length = 7 km

Diagonal Scale

- **Diagonal scale** is **used** in **engineering** to read lengths with **higher accuracy** as it represents into **three** different unit for example meters, centimeters and millimeters.
- When measurements are required up to two decimal place (i.e. **3** unit)
- Example : 6.71 , 7.34 7.34 m $\longrightarrow$ 7 m & 3 dm & 4 cm


For constructing diagonal scale we must know:

1. Its representative fraction i.e. (R.F.) (given or calculate)
2. Maximum length to be measured (given)
3. The units to be shown, for example, millimeters and centimeters (given)

Steps for constructing diagonal scale

Unit 3 ????

1. Calculate the length of scale.

length of scale = R.F X length to measured

2. Draw two horizontal lines parallel to each other and some distance apart so that they give appearance as a rule. Join the ends by vertical lines.

(1. Length for horizontal parallel lines = length of scale which is calculated previously 2. assume suitable length)

3. Divide horizontal lines into required number of equal parts and (depend upon length to measured or read upto)

4. draw perpendicular lines at these division points.

5. Mark 'zero' at the end of first division.

6. sub-divide only first main part to the left of zero into the required number of sub-division according to unit.

1 m = 10 decimeter

Hence we need to divide first part into 10 equal numbers

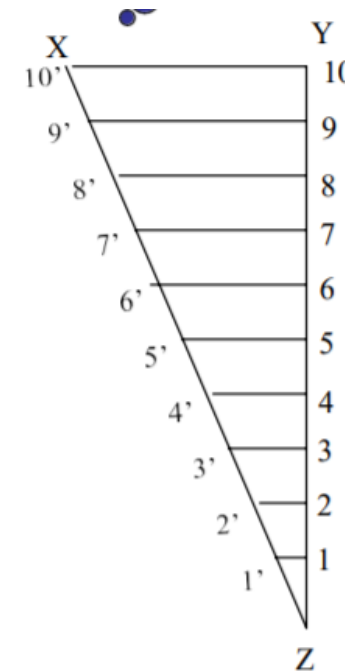
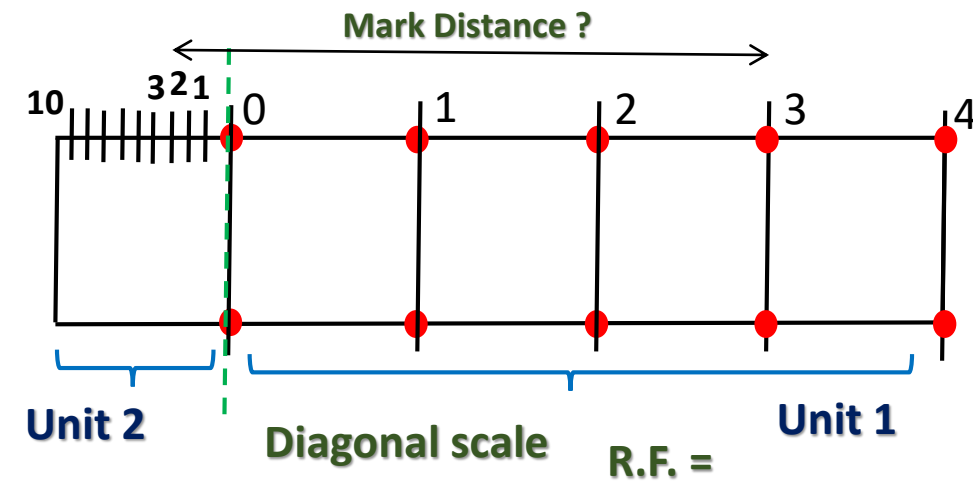
7. Sub-divisions are numbered to the right, starting from zero.

8. Mark the given distance on scale

9. name of the scale and R.F. should be mentioned below scale

After 7 th step we need to perform few different steps to show 3 rd unit

Principle for diagonal scale :



Principle for diagonal scale :

1. Let the XY in figure be a subunit
2. From Y draw a perpendicular YZ to a suitable height. 10
Join XZ. Divide YZ in to 10 equal parts.
3. Draw parallel lines to XY from all these divisions and number them 1', 2' and so on.
4. From geometry we know that similar triangles have their like sides proportional.
5. Consider two similar triangles XYZ and 7'7Z.

$$\Rightarrow \frac{7Z}{YZ} = \frac{7'7}{XY}$$

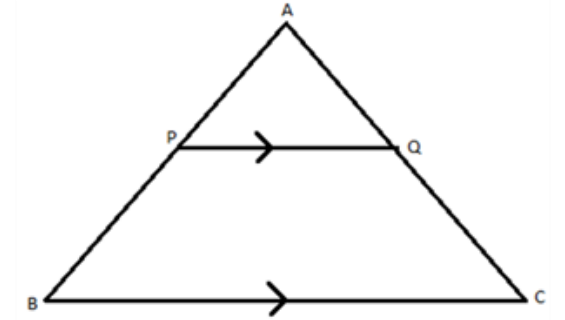
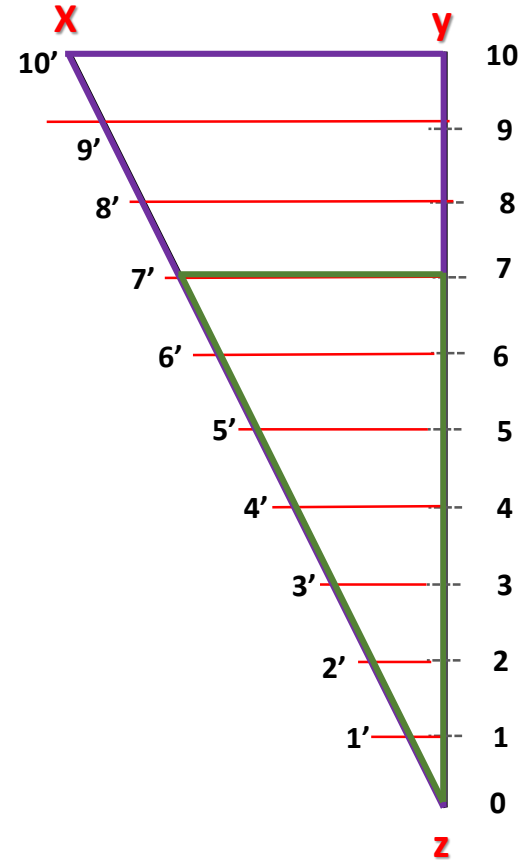
$$\Rightarrow \frac{7 \times XY}{10} = 7'7 \quad \left[7Z = 7 \text{ AND } YZ = 10 \right]$$

$$\Rightarrow \frac{7}{10} \times XY = 7'7$$

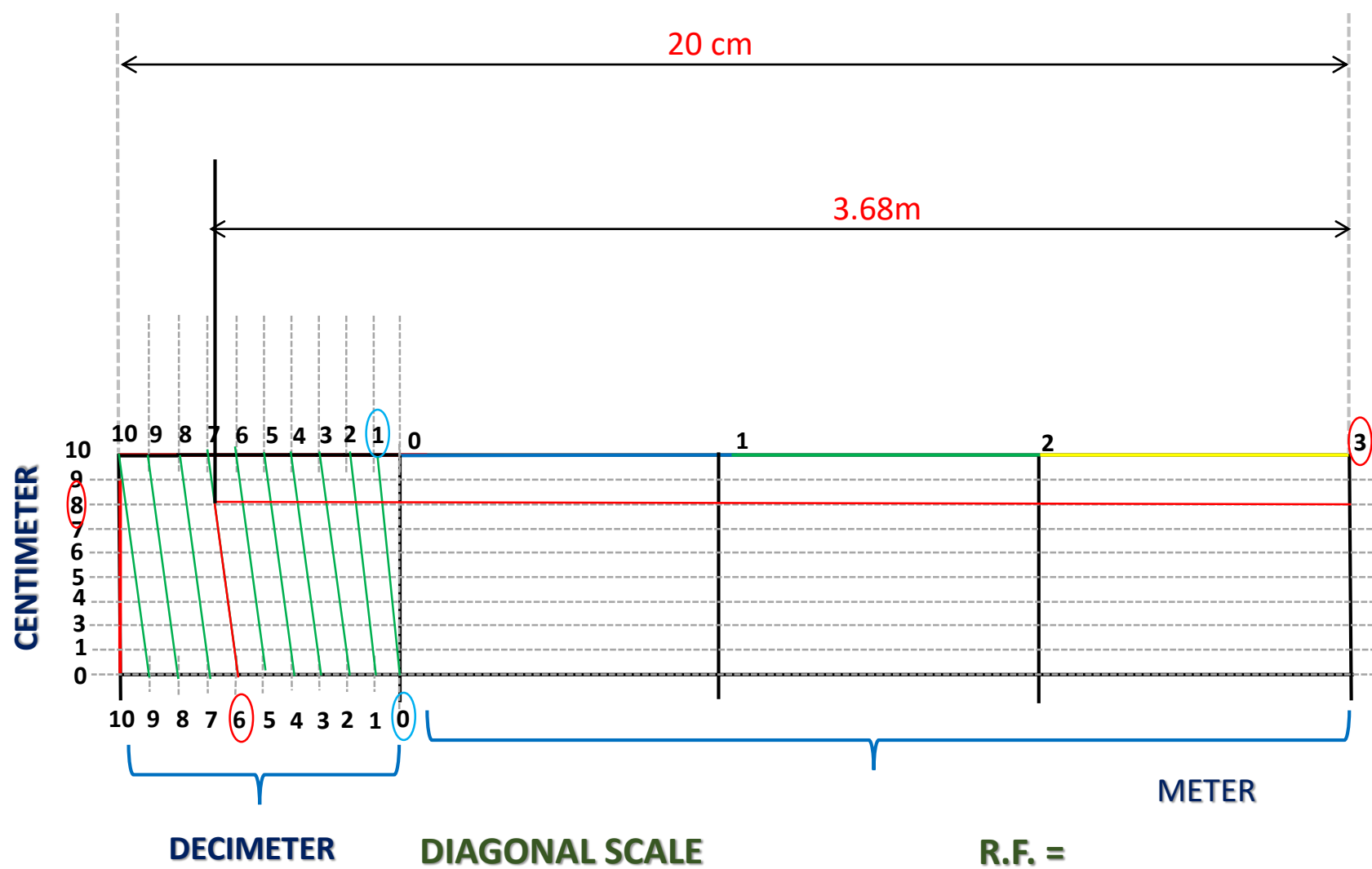
$$\Rightarrow 7'7 = 0.7 \times XY$$

6. Similarly $1' - 1 = 0.1 XY$, $2' - 2 = 0.2 XY$

Thus, it is very clear that, the sides of small triangles, which are parallel to divided lines, become progressively **shorter** in length by **0.1 XY**.



In $\triangle ABC$ and $\triangle APQ$, $\angle PAQ$ is common and $\angle APQ = \angle ABC$ (corresponding angles)
 $\Rightarrow \triangle ABC \sim \triangle APQ$ (AA criterion for similar triangles)
 $\Rightarrow AP/AB = PQ/BC$



Example : 3.48 m means 4 meter and 6 decimeter AND 8 centimeter

Problem 1 :

Construct a diagonal scale of **1:50** to read **meters, decimeters and centimeters** to measure **upto 5 meters**. Show the length of **2 meters, 6 decimeters and 5 centimeters** on the scale

Given data : R.F. = 1/50 , maximum length = 5 m and mark 2.65 m

$$\begin{aligned}\text{length of scale} &= \text{R.F.} \times \text{length to measured} \\ &= 1/50 \times 5 \text{ m} \\ &= 0.1 \text{ m} \\ &= 0.1 \times 100 = \mathbf{10 \text{ cm}}\end{aligned}$$

Problem 2 :

On a drawing, a line **10 cm** long represents a distance of **5 m**. Construct a diagonal scale to **read upto 6 m**, showing **meters ,decimeters and centimeters**. Indicate on your scale the length **4.57 m**

Given data : R.F. = 10 cm /5 m , maximum length = 6 m and mark 4.57 m

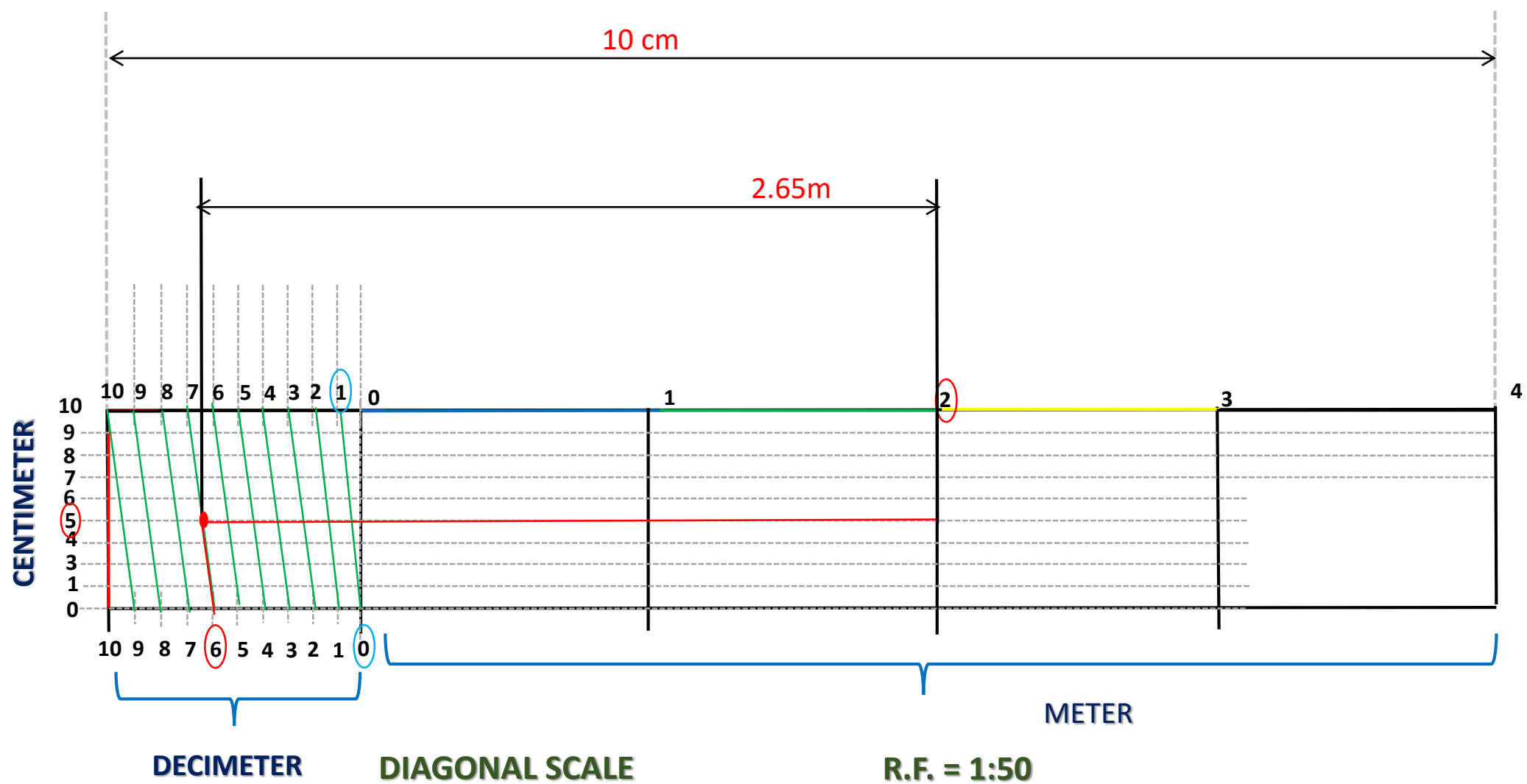
$$\begin{aligned}\text{length of scale} &= \text{R.F.} \times \text{length to measured} \\ &= 1/50 \times 6 \text{ m} \\ &= 0.02 \times 6 \text{ m} \\ &= 0.12 \text{ m} = 0.12 \times 100 \text{ cm} = \mathbf{12 \text{ cm}}\end{aligned}$$

$$\text{R.F.} = \frac{10 \text{ cm}}{5 \text{ m}} = \frac{10 \text{ cm}}{5 \times 100 \text{ cm}} = \frac{10 \text{ cm}}{500 \text{ cm}} = \frac{10}{500} = \frac{1}{50}$$

Problem 3 :

Construct a diagonal scale with R.F. = 1/2000000 to measure upto 300 KM. show on the scale the distance of 236 KM

$$\begin{aligned}\text{R.F.} &= \frac{1}{20,00,000} \times 300 \text{ KM} \\ &= \frac{1}{20,00,000} \times 300 \times 100000 \text{ CM} = \frac{30}{2} = \mathbf{15 \text{ cm}}\end{aligned}$$



Example : 2.65 m means 2 meter and 6 decimeter AND 5 centimeter

Problem 4 :

Construct a diagonal scale with R.F. **1:24** to show **yards, feet** and **inches** and long enough to **measure up to 4 yards**, show a distance of **1 yard** and **2 feet and 4 inches** on your scale.

Given data : R.F. = $1/24$, maximum length = 4 yards

Convert 4 yards into CM



4 yards X 3 (1 foot = 3 yards)



12 feet X 12 (1 foot = 12 inches)



144 inches X 2.54 cm (1 inch = 2.54 cm)



365.76 cm

Hence 4 yards = 365.76 cm

$$\begin{aligned}\text{length of scale} &= \text{R.F} \times \text{length to measured} \\ &= 1/24 \times 365.76 \\ &= \mathbf{15.24 \text{ cm}}\end{aligned}$$